SABIO-RK

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More findability, more interoperability for SABIO-RK, the curated database for biochemical reaction kinetics

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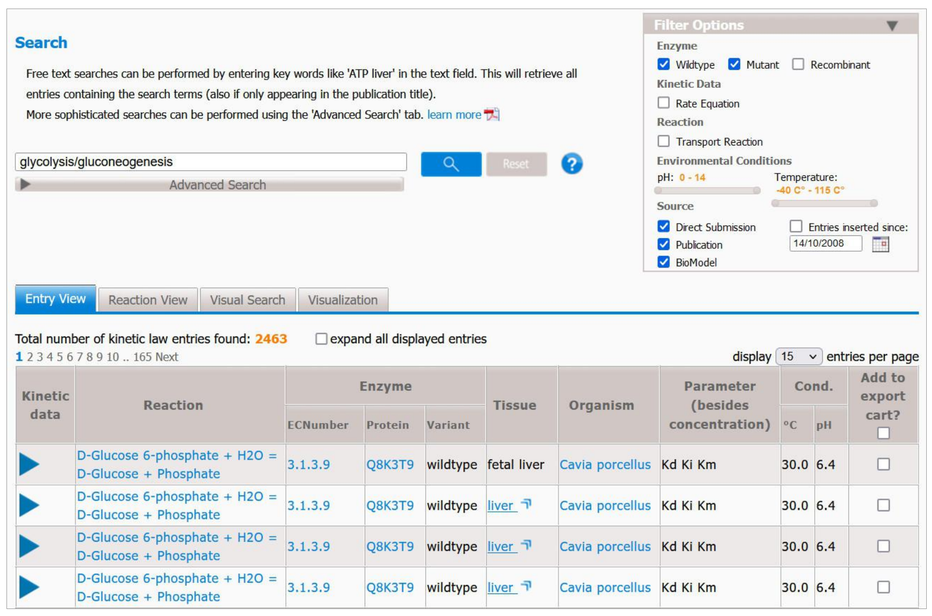
Abstract

SABIO-RK represents a repository for structured, curated, and annotated data on reactions and their kinetics. The data are manually extracted from the scientific literature and stored in a relational database. The content comprises both naturally occurring and alternatively measured biochemical reactions, and the data are made available to the public via a web-based search interface as well as easy-to-use JSON web services for programmatic access.

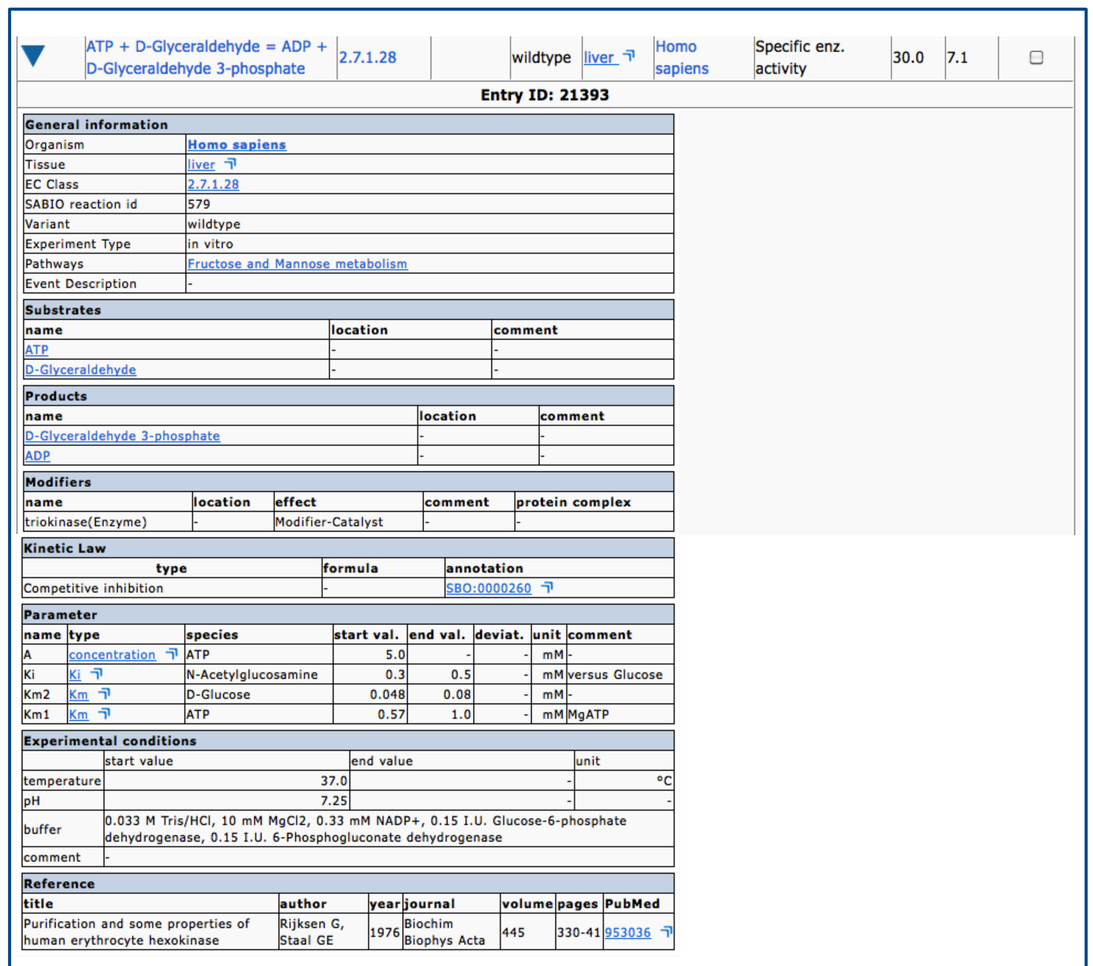
Data are highly interlinked to external databases, ontologies, and controlled vocabularies. This includes cross-references with eg Uniprot, ChEBI, KEGG, BRENDA, Biomodels, and MetaNetX.

In the past year we have worked on improving findability of SABIO-RK data as well as interoperability: SABIO-RK was extended to read the additional annotations in the EnzymeML data exchange format to allow the direct import of enzymology data from EnzymeML documents. SABIO-RK is part of the EnzymeML workflow to support the data transfer between experimental platforms, modelling tools and databases (Range et al. FEBS J 2021).

In the BMBF-funded project SABIO-VIS we focused on visualizing SABIO-RK data for the purpose of interactive search and search refinement.



Search in SABIO-RK: reaction based entries containing kinetics data



Search result details: kinetic parameter values, kinetic law, experimental conditions, organism, tissue and publication

EnzymeML

```
<?xml version="1.0" encoding="UTF-8"?>
<message>upload successfull</message>
<url>http://sabiork.h-its.org/datainput/ViewInfosourceEntries.jsp?editID=7763</url>
```

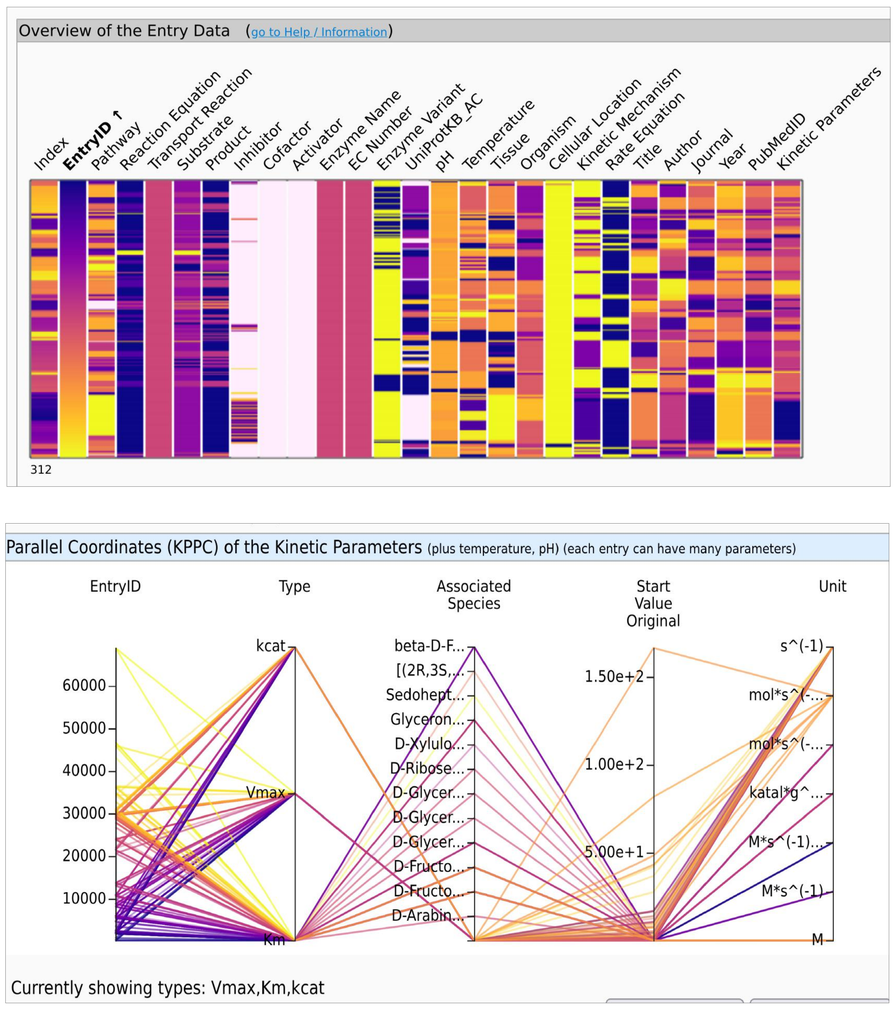
ID	Title	Authors	Journal	Volume	Pages	Year	PubMed	Type	Identifier
7912	Desulfuration of Cysteine	EnzymeML importer	EnzymeML file			2021	not available	Direct submission	https://identifiers.org/doi:10.5281/zenodo.3957403

Upload and data extraction of EnzymeML files

EnzymeML is an XML-based data exchange format that supports the comprehensive documentation of enzymatic data by describing reaction conditions, time courses of substrate and product concentrations, the kinetic model, and the estimated kinetic constants. EnzymeML is based on the Systems Biology Markup Language, which was extended by implementing the STRENDA Guidelines. An EnzymeML document serves as a container to transfer data between experimental platforms, modeling tools, and databases (Range et al. 2021).

The SABIO-RK platform was extended to extract and store data of EnzymeML documents either using manual upload or RESTful web services.

(http://sabiork.h-its.org/datainput/SBMLUpload/uploadEnzymeML)



Interactive visualization of SABIO-RK data

The upper part shows an overview of the search result represented as a heat-map visualization. The lower part shows selected kinetic parameters of the search result represented as a parallel coordinates graph.

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elixir

KTS Klaus Tschira Stiftung gemeinnützige GmbH

de.NBI German Network for Bioinformatics Infrastructure

Bundesministerium für Bildung und Forschung